Atonu Ghosh

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Professional Summary

Final-semester Ph.D. candidate at IIT Kharagpur currently working as a **Research Intern** at **Samsung R&D Institute India - Bangalore**. 5+ years of experience in designing, prototyping, and deploying full-stack IoT and Edge-AI systems. Proven track record of leading projects from concept to production for clients like Tata Steel Downstream Products Limited. Innovator with 15 patents (3 granted, **4 commercialized/in use**) and 8 authored journal articles in secure, scalable IoT architectures, and LPWANs. Dedicated to forward R&D in IoT and edge-intelligence for reliable, human-centric systems.

Education

2021–Present: **Ph.D., Computer Science & Engineering**.
Indian Institute of Technology, Kharagpur, India
Supervisor: Prof. Sudip Misra
IoT, Edge Computing & ML, Low Power Networks, Condition Monitoring, Digital Transformation

2020–2018: **Master of Technology, Computer Science & Engineering**.
Maulana Abul Kalam Azad University of Technology (formerly WBUT), Nadia, West Bengal, India
CGPA: 8.93/10

2014–2017: **Bachelor of Technology, Computer Science & Engineering**.
Institute of Engineering and Management (IEM), Salt Lake, Kolkata, India
CGPA: 7.71/10

Skills

Programming	Python, Micropython, C
Cloud & DevOps	AWS, Azure, Google Cloud, Docker, Kubernetes, Nginx, REST API, OpenStack
IoT & Hardware	Raspberry Pi, ESP32, ESP8266, Raspberry Pi Pico, 8051, LoRa, Sensors, Actuators
Database	SQL, MySQL, PostgreSQL, MariaDB, InfluxDB, SQLite, Redis
ML, Edge,& Tools	Edge ML Integration, YOLO, GitHub, Wireshark, Altium Designer, Fusion 360, Proteus, Keil μVision, NodeRed

Professional & Research Experience

Jan 2026 – Present: **Research Intern**, **Samsung R&D Institute India - Bangalore**.

Jan 2021 – Present: **Ph.D. Research Scholar**, **Indian Institute of Technology, Kharagpur**.

- **EOT Crane Condition Monitoring at Tata Steel Downstream Products Ltd.**
  Led a team of 4 to deploy a **full-stack, production** system. Architected the system using Python, MQTT, **Docker**, PostgreSQL, Redis, and **Azure** for real-time analytics.
- **TLoRa: TLS 1.3 Handshake over LoRa Backhaul**
  Engineered a novel transparent proxy establishing a **full end-to-end TLS 1.3 handshake over LoRa**, enabling secure HTTPS for ultra-low-power, long-range IoT devices.
- **LoRaConnect: HTTP-over-LoRa Application Proxy**

Developed an application-layer proxy for HTTP-over-LoRa, implementing message chunking, reassembly, and FHSS to overcome LoRa payload limitations.

- **Production Server Management & DevOps (MeitY Project)**
  Managed & deployed production web services for a **Govt. of India (MeitY)** project. Containerized with **Docker**, managed **Nginx** reverse proxies, & built Django dashboard.

- **Resilient IoT Data Infrastructure**
  Implemented **automated SQL (MariaDB) database replication** to ensure data integrity from remote IoT nodes with intermittent network connectivity.

Nov 2020 – Oct 2025 **System Architect, *SensorDrops Networks Pvt. Ltd., IIT Kharagpur***.
Contributed voluntarily without pay to this startup founded by my Ph.D. supervisor.

- **Pet Immune Human Intruder Detection and Alert System**
  Architected system to integrate with CCTV DVRs & led a team of 3 to develop & deploy an Edge ML system for real-time pet immune human intruder detection.

- **Machine and Ambient Condition Monitoring System**
  Redesigned the existing system to include electrical parameters along with vibration and temperature monitoring of electrical motors. Modularized and made the system extensible.

- **Smartphone Controller for Electrical Lights and Fans**
  Managed a team of 3 to develop a system that connects with existing electrical wiring and enables the existing lights and fans to be controlled using a smartphone application.

Patents & Publications

Selected Patents (Granted & Commercialized)

P3 Sudip Misra, **Atonu Ghosh,** Bontalakoti Aditya, Kumarjit Ray, and Rajdeep Ghosh, An Intelligent Liquid-Level Sensing and Pump Control System, *Indian Patent*, File No. – 202531098034, Oct 10, 2025**(Commercialized)**.

P2 **Atonu Ghosh,** Biswajit Ghosh, Ruelia Saha, Sudip Misra, and Arijit Roy, System for Real Time Intrusion Detection, Actuation and Alert and Method Thereof, *Indian Patent*, Number – 566326, May 16, 2025 **(Commercialized)**.

P1 Sudip Misra, **Atonu Ghosh,** Biswajit Ghosh, Swaraj Lahiri, Bharat Bhushan, Subrata Ray, and Sonali Pan, Embedded System and Method for Condition Monitoring of Electric Overhead Travelling (EOT) Cranes, *Indian Patent*, File No. – 202531033456, Apr 4, 2025 **(In Use at Tata Steel Downstream Products Ltd.)**.

Selected Publications (Published, Preprints, and Communicated)

J3 **Atonu Ghosh,** Akhilesh Mohanasundaram, Srishivanth R F, and Sudip Misra, TLoRa: Implementing TLS Over LoRa for Secure HTTP Communication in IoT, In *IEEE Transactions on Industrial Informatics, 2025.*

J2 **Atonu Ghosh,** David Breuss, Simon Howind, Sudip Misra, Thilo Sauter, and Surjya Kanta Pal, SenseAI: Staged Model Deployment for Resource Constrained AIoT-Enabled Industrial Environments, In *IEEE Transactions on Industrial Cyber-Physical Systems, 2025.*

J1 **Atonu Ghosh,** Anandarup Mukherjee, and Sudip Misra, SEGA: Secured Edge Gateway Microservices Architecture for IIoT-based Machine Monitoring, In *Transactions on Industrial Informatics, 2021.*

Achievements & Activities

2024 **Qualcomm Innovation Fellowship**, Finalist

2022-2023 **Secretary of IEEE Student Branch Chapter**, IIT Kharagpur